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JEE Mains Maths Formulas

Check Important Formulas List

Topic - 1: Logarithms

- $log\ (x * y) = log\ x + log\ y$
- $log\ (x/y) = log\ x - log\ y$
- $log\ x^y = y * log\ x$
- $log_a\ b = 1/log_b\ a$
- $log_b\ a = log_c\ a/log_c\ b \rightarrow Base\ Change$
- $a^{log_a x} = x$

Topic - 2: Trigonometry

- $\pi\ Radians = 180^o = 200\ Grade$
- $sin^2\ \theta + cos^2\ \theta = 1$
- $sec^2\ \theta = 1 + tan^2\ \theta$
- $csc^2\ \theta = 1 + cot^2\ \theta$
- $sin\ \theta \geq 0$ in the first and second quadrant.
- $cos\ \theta \geq 0$ in the first and fourth quadrant.
- $tan\ \theta \geq 0$ in the first and third quadrant.
- $sin\ (\ 2n\pi + \theta\ ) = sin\ \theta$ and $sin\ [\ (2n+1)\pi + \theta\ ] = -sin\ \theta$
- $cos\ (\ 2n\pi + \theta\ ) = cos\ \theta$ and $cos\ [\ (2n+1)\pi + \theta\ ] = -cos\ \theta$
- $sin\ (\ 2n\pi - \theta\ ) = -sin\ \theta$ and $sin\ [\ (2n+1)\pi - \theta\ ] = sin\ \theta$
- $sin\ [\ (\pi/2) \pm \theta\ ] = cos\ \theta$
- $cos\ [\ (\pi/2) \pm \theta\ ] = \mp sin\ \theta$
- $sin\ (n * \pi) = 0$ and $cos\ [\ (2n+1)\pi/2\ ] = 0$
- $sin\ (\theta \pm \varphi) = sin\ \theta * cos\ \varphi \pm cos\ \theta * sin\ \varphi$
- $cos\ (\theta \pm \varphi) = cos\ \theta * cos\ \varphi \mp sin\ \theta * sin\ \varphi$
- $tan\ (\theta \pm \varphi) = [\ tan\ \theta \pm tan\ \varphi\ ]/[\ 1 \mp tan\ \theta * tan\ \varphi\ ]$
- $sin\ \theta \pm sin\ \phi = 2 * sin\ [\ (\theta \pm \phi)/2\ ] * cos\ [\ (\theta \mp \phi)/2\ ]$
- $cos\ \theta + cos\ \phi = 2 * cos\ [\ (\theta + \phi)/2\ ] * cos\ [\ (\theta - \phi)/2\ ]$
- $cos\ \theta - cos\ \phi = 2 * sin\ [\ (\theta + \phi)/2\ ] * sin\ [\ (\phi - \theta)/2\ ]$
- $2 * sin\ \theta * cos\ \phi = sin\ (\theta + \phi) + sin\ (\theta - \phi)$
- $2 * cos\ \alpha * cos\ \beta = cos\ (\alpha + \beta) + cos\ (\alpha - \beta)$
- $2 * sin\ \alpha * sin\ \beta = cos\ (\alpha - \beta) - cos\ (\alpha + \beta)$
- $sin\ (2 * \theta) = 2 * sin\ \theta * cos\ \theta$
- $cos\ (2 * \theta) = cos^2\ \theta - sin^2\ \theta = 2 * cos^2\ \theta - 1 = 1 - 2 * sin^2\ \theta$
- $sin\ (3 * \theta) = 3 * sin\ \theta - 4 * sin^2\ \theta$
- $cos\ (3 * \theta) = 4 * cos^3\ \theta - 3 * cos\ \theta$
- $tan\ (3 * \theta) = (\ 3 * tan\ \theta - tan^3\ \theta\ )/(1 - 3 * tan^2\ \theta)$
- $sin^2\ \theta - sin^2\ \phi = sin\ (\theta + \phi) * sin\ (\theta - \phi) = cos^2\ \phi - cos^2\ \theta$
- $cos^2\ \theta - sin^2\ \varphi = cos\ (\theta + \varphi) * cos\ (\theta - \varphi)$
- $\sum_{k=0}^{n-1} \quad sin\ (\alpha + k * \beta) = \dfrac{sin[\ \alpha + \{\ (n-1)/2\ \} * \beta\ ]}{sin(\beta/2)} * sin\ (n\beta/2)$

- $\sum_{k=0}^{n-1} \quad cos\ (\alpha + k * \beta) = \frac{cos[\ \alpha + \{\ (n-1)/2\ \} * \beta\ ]}{sin(\beta/2)} *sin\ (n\beta/2)$
- $sin\ A = sin\ B \Rightarrow A = n\pi + (-1)^n * B$
- $cos\ A = cos\ B \Rightarrow A = 2n\pi \pm B$
- $tan\ A = tan\ B \Rightarrow A = n\pi + B$

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Topic - 3: Inverse Trigonometric Function

- Domain and Range of Inverse Trigonometric Functions -
  - $x \in [\ -1, 1\ ]\ and\ \ arcsin\ x \in [\ -\pi/2, \pi/2\ ]$
  - $x \in [\ -1, 1\ ]\ and\ \ arccos\ x \in [\ 0, \pi\ ]$
  - $x \in (-\infty, \infty)\ and\ \ arctan\ x \in (\ -\pi/2, \pi/2\ )$
- Properties of Inverse Trigonometric Functions -
  - $sin\ (arcsin\ x) = x$
  - $cos\ (arccos\ x) = x$
  - $tan\ (arctan\ x) = x$
  - $csc^{-1}\ x = arcsin\ (1/x)\ for\ |x| \geq 1$
  - $sec^{-1}\ x = cos\ (1/x)\ for\ |x \geq 1|$
  - $cot^{-1}\ x = tan^{-1}\ (1/x)\ for\ x > 0\ and\ \ cot^{-1}\ x = \pi + tan^{-1}\ x\ for\ x < 0$
  - $sin^{-1}\ (-x) = \ -sin^{-1}\ x$
  - $cos^{-1}\ (-x) = \pi - cos^{-1}\ x$
  - $tan^{-1}\ (-x) = \ -tan^{-1}\ x$
  - $cot^{-1}\ (-x) = \pi - cot^{-1}\ x$
  - $sin^{-1}\ x + cos^{-1}\ x = \pi/2$
  - $tan^{-1}\ x + cot^{-1}\ x = \pi/2$
  - $csc^{-1}\ x + sec^{-1}\ x = \pi/2$
  - $tan^{-1}\ x + tan^{-1}\ y = tan^{-1}\ [\ (x + y) \div (1 - xy)\ ]\ when\ x > 0, y > 0\ and\ 0 < xy < 1$
  - $tan^{-1}\ x - tan^{-1}\ y = tan^{-1}\ [\ (x - y) \div (1 + xy)\ ]\ when\ x > 0, y > 0$
  - $sin^{-1}\ x - sin^{-1}\ y = sin^{-1}\ [\ x\sqrt{1 - y^2} - y\sqrt{1 - x^2}\ ]\ for\ x > 0, y > 0\ and\ x^2 + y^2 < 1$
  - $cos^{-1}\ x - cos^{-1}\ y = cos^{-1}\ [\ xy - \sqrt{1 - x^2}\sqrt{1 - y^2}\ ]\ for\ x < y$
  - $sin^{-1}\ [\ 2x/(1 + x^2)\ ] = 2\ tan^{-1}\ x\ \ when\ |x| \leq 1$
  - $cos^{-1}\ [\ (1 - x^2)/(1 + x^2)\ ] = 2\ tan^{-1}\ x\ \ when\ x \geq 0$
  - $tan^{-1}\ [\ 2x/(1 - x^2)\ ] = 2\ tan^{-1}\ x\ \ when\ |x| < 1$
  - $sin^{-1}\ (3x - 4x^3) = 3\ sin^{-1}\ x\ \ when\ |x| \leq 1/2$
  - $cos^{-1}\ (4x^3 - 3x) = 3\ cos^{-1}\ x\ \ when\ 1/2 \leq x \leq 1$
  - $sin^{-1}\ (\ 2x\sqrt{1 - x^2}\ ) = 2\ sin^{-1}\ x\ \ when\ |x| \leq 1/\sqrt{2}$
  - $cos^{-1}\ (\ 2x^2 - 1\ ) = 2\ cos^{-1}\ x\ \ when\ 0 \leq x \leq 1$

Topic - 4: Quadratic Equation

- $ax^2 + bx + c = 0 \Rightarrow x = (1/2a)(-b \pm D)$ where $D = \sqrt{b^2 - 4ac}$
- If α and β are the roots of $f(x) = ax^2 + bx + c$, then $\alpha + \beta = \ -b/a$ and $\alpha * \beta = c/a$
- $D > 0 \Rightarrow$ Roots are real
- $D = 0 \Rightarrow$ Roots are real and identical
- $D < 0 \Rightarrow$ Roots are Imaginary

- If $a + ib$ is one root then $a - ib$ must be the other root.
- Roots of $ax^2 + bx + c = 0$ and $\alpha x^2 + \beta x + \gamma = 0$ then $\frac{a}{\alpha} = \frac{b}{\beta} = \frac{c}{\gamma}$
- If $a, b \geq 0$ then $a + b \geq 2 * \sqrt{a * b}$
- If $f(x) = ax^2 + bx + c$ and $a > 0$ then $f(x) \geq (4ac - b^2)/4a$
- If $f(x) = ax^2 + bx + c$ and $a < 0$ then $f(x) \leq (4ac - b^2)/4a$
- $f(x, y) = ax^2 + by^2 + 2hxy + 2gx + 2fy + c$ can be resolved into two linear factors if and only if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Topic - 5: Sequence and Series

- If the first term is a and common difference is d then the nth $term$ of AP is $a_n = a + (n - 1) * d$
- $\sum_{k=1}^{n} \quad a + (k - 1) * d = \frac{n}{2}[\,2a + (n - 1)d\,] = \frac{n}{2}(a + a_n)$
- If $a_n = An + B$ then $a = A + B$ and $d = B$
- If $S_n = An^2 + Bn$ then $a = A + B$ and $d = 2 * A$
- The Arithmetic Mean of a and b is $(a + b) / 2$
- If $a, A_1, A_2, A_3, \ldots, A_n, b$ are in AP with A_k $(k = 1,2,3,\ldots,n)$ is the Arithmetic Mean, then the common difference is $d = (b - a)/(n + 1)$
- If the first term is a and the common ratio is r then the nth $term$ of GP is $a_n = a * r^{n-1}$
- $\sum_{k=1}^{n} \quad a * r^{k-1} = \frac{a * (r^n - 1)}{r - 1}$ where $r \neq 1$
- $\sum_{k=1}^{\infty} \quad a * r^{k-1} = \frac{a}{1 - r}$ where $|r| < 1$
- The Geometric Mean of a and b is $\sqrt{a * b}$
- If $a, G_1, G_2, G_3, \ldots, G_n, b$ are in GP with G_k $(k = 1,2,3,\ldots,n)$ are the Geometirc Mean, then the common ratio is $r = (b/a)^{1/(n+1)}$
- If $a_1, a_2, a_3, \ldots, a_n > 0$ are terms in GP then $log\ a_1, log\ a_2, log\ a_3, \ldots, log\ a_n$ are in AP
- If the first term is a and the common difference is d then the nth $term$ of HP is $a_n = 1 / [\,a + (n - 1) * d\,]$
- The Harmonic Mean of a and b is $2ab/(a + b)$
- $AM \geq GM \geq HM$
- $\sum_{k=1}^{n} \quad k = \frac{n(n+1)}{2}$, $\sum_{k=1}^{n} \quad k^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum_{k=1}^{n} \quad k^3 = \frac{n^2(n+1)^2}{4}$

Topic - 6: Binomial Theorem

- $(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \ldots + {}^nC_n y^n$
- The $(r + 1)^{th}$ $term$ of $(x + y)^n$ is $T_{r+1} = {}^nC_r x^{n-r} y^r$
- $C_x = C_y \Rightarrow x = y\ or\ x + y = n$
- $C_{r-1} + C_r = {}^{n+1}C_r$
- $\sum_{k=0}^{n} \quad \frac{C_k}{k+1} = \frac{2^{n+1} - 1}{n+1}$ and $\sum_{k=0}^{n} \quad (-1)^k \frac{C_k}{k+1} = \frac{1}{n+1}$
- $C_0 + C_1 + C_2 + \ldots + C_n = 2^n$ and ${C_0}^2 + {C_1}^2 + {C_2}^2 + \ldots + {C_n}^2 = {}^{2n}C_n$

- The greatest coefficient of $(x+y)^n$ is $C_{n/2}$ is n is even or $C_{(n-1)/2}$ and $C_{(n+1)/2}$ if n is odd.
- $(1+x)^k = 1 + kx + k(k+1)x^2/2! + k(k+1)(k+2)x^3/3! + \ldots \infty$ if $k \in R$
- $(1-x)^{-1} = 1 + x + x^2 + x^3 + \ldots \infty$
- $(1+x)^{-1} = 1 - x + x^2 - x^3 + \ldots \infty$
- $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ and $ln\,(1+x) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k}$ where $x \neq -1$

Topic - 7: Complex Numbers

- If $z = a + ib$ then $\underline{z} = a - ib$
- $z + \underline{z} = 2 * Re(z)$, $z - \underline{z} = 2 * i\, Im(z)$ and $z\underline{z} = a^2 + b^2$
- $|z| = \sqrt{a^2 + b^2}$ and $Arg(z) = tan^{-1}\,(y/x)$
- $Arg(z_1 * z_2) = Arg(z_1) + Arg(z_2)$, $Arg(z_1/z_2) = Arg(z_1) - Arg(z_2)$ and $Arg(z^n) = n * Arg(z)$
- $z = r(cos\,\theta + i\,sin\,\theta) = re^{i\theta}$ where r is the magnitude and θ is the argument.
- Euler's Formula is $e^{i\theta} = cos\,\theta + i\,sin\,\theta$
- $2\,cos\,\theta = e^{i\theta} + e^{-i\theta}$ and $2\,sin\,\theta = e^{i\theta} - e^{-i\theta}$
- If $f(\alpha + i\beta) = a + ib$ then $f(\alpha - i\beta) = a - ib$
- The triangle inequalities are $|\,z_1 + z_2\,| \leq |\,z_1\,| + |\,z_2\,|$ and $|\,z_1 - z_2\,| \geq |\,|\,z_1\,| - |\,z_2\,|\,|$
- If $z = r(cos\,\theta + i\,sin\,\theta)$ then $z^n = r^n[\,cos\,(n\theta) + i\,sin\,(n\theta)\,]$, this is De Moiver's Theorem
- Cube Root of Unity is $1, \omega\ and\ \omega^2$ where $\omega = \frac{-1 + i\sqrt{3}}{2}$ and $\omega^2 = \frac{-1 - i\sqrt{3}}{2}$
- $1 * \omega * \omega^2 = 1$ and $1 + \omega + \omega^2 = 0$
- $a^2 + b^2 + c^2 - ab - bc - ca = (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$
- $a^3 + b^3 = (a+b)(a\omega + b\omega^2)(a\omega^2 + b\omega)$
- $x^2 + x + 1 = (x - \omega)(x - \omega^2)$
- If $z = r(cos\,\theta + i\,sin\,\theta)$ then $z^{1/n} = r^{1/n}[\,cos\,\{(\theta + 2\pi k)/n\} + i\,sin\,\{(\theta + 2\pi k)/n\}\,]$
- Equation of a Circle with center z_0 and radius r is $|\,z - z_0\,| = r$

Topic - 8: Matrices

- Sum of diagonal elements or the Trace of a Matrix is $tr(A) = \sum_{i=1}^{n} a_{ii}$.
- Two Matrices $[a_{ij}] = [b_{ij}]$ are equal then $a_{ij} = b_{ij}$.
- $[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}]$ and $k * [a_{ij}] = [k * a_{ij}]$.
- Define $A_{m\times n} = [a_{ij}]$ and $B_{n\times p} = [b_{ij}]$, then $AB = [\sum_{k=1}^{n} a_{ik} \cdot b_{kj}]$.
- Cayley-Hamilton's Theorem: $|\,A - \lambda I\,| = 0$ where λ is the Eigenvalue.
- Commutative Property of Matrix Addition → $A + B = B + A$.
- Associative Property of Matrix Addition → $(A + B) + C = A + (B + C)$.
- Associative Property of Matrix Multiplication → $(AB)C = A(BC)$.
- Distributive Property of Matrices → $A(B + C) = AB + AC\ and\ (B + C)A = BA + CA$.
- Define $n, m \in N$ then $A^n \cdot A^m = A^{n+m}$, $(A^n)^m = A^{nm}$ and $I^n = I$ where I is the Identity Matrix.
- Orthogonal Matrix → $A^{-1} = A^T\ or\ A^T A = I$.
- Symmetric Matrix → $A^T = A$ and Skew-Symmetric Matrix $A^T = -A$.
- The diagonal elements of a Skew-Symmetric Matrix are 0, that is $a_{ii} = 0$.
- $A(adjA) = (adjA)A = |A|\,I_n \Rightarrow |\,adjA\,| = |\,A\,|^{n-1}$.
- $adj(adjA) = |\,A\,|^{n-2}A$ and $|\,adj(adjA)\,| = |\,A\,|^{(n-1)(n-1)}$.

- $adj(AB) = (adjB)(adjA)$ and $adj(kA) = k^{n-1}adjA$ where k is a constant.
- The Inverse of a Matrix is $A^{-1} = \frac{1}{|A|} adjA$, the inverse exists only when $|A| \neq 0$.
- Inverse of a Matrix Product is $(AB)^{-1} = B^{-1}A^{-1}$.
- The solution to the system of equations $AX = B$ is $X = \frac{adjA}{|A|} B$ where $|A| \neq 0$.
- If $|A| \neq 0$ but $(adjA)B = 0$ then the solution is $x = y = z = 0$.
- If $|A| = 0$ and $(adjA)B = 0$ then the system of equations has Infinite Solutions.
- If $|A| = 0$ and $(adjA)B \neq 0$ then there is NO Solution.

Topic - 9: Determinants

- For an $n \times n$ Determinant, if a_k is the $k\ th$ element in i^{th} row with Cofactor A_K, then the value of hte Determinant is $\sum_{k=1}^{n} \quad a_k A_k$.
- The value of the Determinant does not change if the rows and columns are interchanged.
- Interchanging the $i\ th$ row with $(i \pm 1)\ th$ row will change the sign of the Determinant.
- If any two rows (or columns) are proportional then the Determinant evaluates to Zero.
- $$\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$
- Row Operation → $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 + ma_1 & b_2 + mb_1 & c_2 + mc_1 \\ a_3 + na_1 & b_3 + nb_1 & c_3 + nc_1 \end{vmatrix}$.
- If the Determinant is $$D(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$$ then its derivative is written as

$$\frac{dD}{dx} = \begin{vmatrix} \frac{d}{dx} f_1(x) & \frac{d}{dx} f_2(x) & \frac{d}{dx} f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ \frac{d}{dx} g_1(x) & \frac{d}{dx} g_2(x) & \frac{d}{dx} g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$$
$$+ \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ \frac{d}{dx} h_1(x) & \frac{d}{dx} h_2(x) & \frac{d}{dx} h_3(x) \end{vmatrix}.$$

- Multiplication of Determinants → $|[a_{ij}]| * |[b_{ij}]| = |[a_{ij} * b_{ij}]|$.
- Determinant of a Skew-Symmetric Matrix is Zero.

- $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a) = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix}.$
- $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$
- $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2+b^2+c^2-ab-bc-ca)$
- If we have $a_k x + b_k y + c_k = d_k \; \forall \; k = 1,2,3$ then the solution using the Cramer's Rule is $x = \Delta_1 / \Delta$, $y = \Delta_2 / \Delta$ and $z = \Delta_3 / \Delta$ where $\Delta_1, \Delta_2, \Delta_3 \; and \; \Delta$ are as shown

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \quad \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}.$$

Topi - 10: Properties and Solutions of a Triangle

- The Law of Sine is $\frac{a}{sinA} = \frac{b}{sinB} = \frac{c}{sinC}$
- The Law of Cosine is $cos\ A = \frac{b^2+c^2-a^2}{2bc}$
- The Law of Tangent is $tan\ [\ (A-B)/2\ ] = \frac{a-b}{a+b}\ cot\ (A/2)$
- The Projection Formula is $a = b\ cos\ C + c\ cos\ B$
- $sin\ (A/2) = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $cos\ (A/2) = \sqrt{\frac{s(s-a)}{bc}}$ and $tan\ (A/2) = \frac{\Delta}{s(s-a)}$ where $s = \frac{a+b+c}{2}$
- Area of a Triangle is $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$
- Area of an Equilateral Triangle is $\Delta_{eq} = \frac{\sqrt{3}}{4} l^2$ where l is the length of each side of the triangle.
- Area of a Triangle given an angle is $\Delta = \frac{1}{2} bc\ sin\ A = \frac{1}{2} ca\ sin\ B = \frac{1}{2} ab\ sin\ C$

- Radius of the Circumcenter is $R = \frac{a}{2sinA} = \frac{b}{2sinB} = \frac{c}{2sinC} = \frac{abc}{4\Delta}$
- Radius of Incircle is $r = \Delta/s = (s-a)\ tan\ (A/2) = (s-b)\ tan\ (B/2) = (s-c)\ tan\ (C/2)$
- Radius of Excircle is $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$ and $r_3 = \frac{\Delta}{s-c}$
- Length of Median from angle A is $m_A = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$
- Length of Angle Bisector of A is $b_A = \frac{2bc}{b+c}\ cos\ (A/2)$
- Length of Altitude from A is $h_A = \frac{a}{cotB + cotC}$
- ${m_A}^2 + {b_A}^2 + {h_A}^2 = \frac{3}{4}(\ a^2 + b^2 + c^2\ )$
- For an Isosceles Triangle $a\ cos\ B = b\ cos\ A$
- $r = \frac{a}{2}\ cot\ (\pi/n)$ and $R = \frac{a}{2}\ csc\ (\pi/n)$ for a Regular Polygon with n sides and side length a
- Area of a Cyclic Quadrilateral with sides $a, b\ , c\ and\ d$ is $\Delta = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ where $s = (a+b+c+d)/2$
- Ptolemy's Theorem is $AC * BD = a * c + b * d$

Topic - 11: Straight Lines

- $x = r\ cos\ \theta$ and $y = r\ sin\ \theta$ where $r = \sqrt{x^2 + y^2}$ and $\theta = tan^{-1}\ (y/x)$
- Distance Formula in 2-D is $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- Coordinates for Internal Division is $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
- Coordinates for External Division is $\left(\frac{mx_2-nx_1}{m-n}, \frac{my_2-ny_1}{m-n}\right)$
- Coordinates of the Centroid is $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$
- Coordinates of Incenter is $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$
- Area of a Triangle with vertices $(x_1,\ y_1)$, (x_2, y_2) and (x_3, y_3) is $\left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

  Latex Code: $\left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

- Equation of $x - axis$ is $y = 0$ and the Equation of $y - axis$ is $x = 0$
- Slope of a line is given as $m = \frac{y_2-y_1}{x_2-x_1}$
- Slope Intercept Form of a Line: $y = mx + c$
- Point Slope Form of a Line: $y - y_1 = m(x - x_1)$

- Two Point Form of a Line: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
- Intercept Form of a Line: $\frac{x}{a} + \frac{y}{b} = 1$
- Normal Form of a Line: $x \cos\alpha + y \sin\alpha = p$
- Parametric Form of a Line: $x = h + r\cos\theta$ and $y = k + r\sin\theta$
- General Form of a Line: $ax + by + c = 0$
- Angle between two lines is $\theta = \pm \tan^{-1}\left(\frac{m_1 - m_2}{1 + m_1 m_2}\right)$
- Lines are Parallel if and only if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
- Lines are Perpendicular if and only if $a_1 a_2 + b_1 b_2 = 0$
- Lines are Coincident if and only if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- Lines are Intersecting if and only if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
- Length of Perpendicular from (x_1, y_1) to the line $ax + by + c = 0$ is $\left|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$
- Distance between two Parallel Lines is $\left|\frac{c_1 - c_2}{\sqrt{a^2 + b^2}}\right|$
- Line parallel to $ax + by + c = 0$ has the equation $ax + by + k = 0$ where $c \neq k$
- Line perpendicular to $ax + by + c = 0$ is $bx - ay + \lambda = 0$
- Three lines of the form $ax + by + c = 0$ are concurrent if and only if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$
- Linear Translation of axes by (α, β): $x_{new} = x - \alpha$ and $y_{new} = y - \beta$
- Rotation of axes by angle θ anti-clockwise: $x_{new} = x\cos\theta + y\sin\theta$ and $y_{new} = -x\sin\theta + y\cos\theta$
- Equation of the line passing through the intersection of $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$ is $a_1 x + b_1 y + c_1 + \lambda(a_2 x + b_2 y + c_2) = 0$
- $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is the equation for pair of straight lines if and only if $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$ or $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$
- For a pair of straight line $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, angle between them is $\theta = \pm \tan^{-1}\left(\frac{2\sqrt{h^2 - ab}}{a + b}\right)$. Lines are Perpendicular if $a + b = 0$
- Equation of the Angle Bisectors of $ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$
- Pair of Straight Lines perpendicular to $ax^2 + 2hxy + by^2 = 0$ is $bx^2 - 2hxy + ay^2 = 0$

- Coordinates of the foot of the perpendicular (h, k) from (x_1, y_1) to the line $ax + by + c = 0$ is $\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{ax_1+by_1+c}{a^2+b^2}$
- Coordinate (h, k) of the image of a point (x_1, y_1) over the line $ax + by + c = 0$ is $\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{2*(ax_1+by_1+c)}{a^2+b^2}$

Topic - 12: Circle

- Standard Equation of a Circle is $(x-h)^2 + (y-k)^2 = r^2$ where the centre is (h, k) with Radius r.
- General Equation of a Circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ where the coordinates of the centre is $(-g, -f)$ with Radius $\sqrt{g^2 + f^2 - c}$.
- If $g^2 + f^2 - c = 0$ then the Circle is actually the point $(-g, -f)$.
- If $g^2 + f^2 - c < 0$ then the Circle is Imaginary.
- If the end-points of the Diameter are (x_1, y_1) and (x_2, y_2), then the Equation of the Circle is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.
- The parametric form of $(x-h)^2 + (y-k)^2 = r^2$ is $x = h + r\cos\theta$ and $y = k + r\sin\theta$.
- The parametric form of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $x = -g + \cos\theta \cdot \sqrt{g^2 + f^2 - c}$ and $y = -k + \sin\theta \cdot \sqrt{g^2 + f^2 - c}$.
- The point $(r\cos\alpha, r\sin\alpha)$ is simply written as point α.
- The equation of a line joining the points α and β of the circle $x^2 + y^2 = a^2$ is $x\cos\{(\alpha+\beta)/2\} + y\sin\{(\alpha+\beta)/2\} = a\cos\{(\alpha-\beta)/2\}$.
- Define $C(x, y) = x^2 + y^2 + 2gx + 2fy + c$ and (a, b) is any arbitrary point, then -
  - If $C(a, b) = 0 \Rightarrow$ The point is on the boundary of the Circle.
  - If $C(a, b) < 0 \Rightarrow$ The point lies inside the Circle.
  - If $C(a, b) > 0 \Rightarrow$ The point lies outside the Circle.
- For $x^2 + y^2 = r^2$, the Equation of the Tangent at (a, b) is $ax + by = r^2$.
- For $x^2 + y^2 = r^2$, the Equation of the Tangent at $(r\cos\theta, r\sin\theta)$ is $x\cos\theta + y\sin\theta = r$.
- Two Tangents at points α and β of the Circle $x^2 + y^2 = r^2$ intersects each other at the point
$$\left(\frac{r\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{r\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$$
- The Equation of Tangent for the Circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at the point (a, b) is $ax + by + g(x+a) + f(y+b) + c = 0$.
- Equation of the Tangent on the Circle $x^2 + y^2 = r^2$ with Slope m is $y = mx \pm r \cdot \sqrt{1+m^2}$.
- Equation of the Tangent on the Circle $(x-h)^2 + (y-k)^2 = r^2$ with Slope m is written as $y - k = (x-h) \pm r \cdot \sqrt{1+m^2}$.
- The length of Tangent from (a, b) to the Circle $C(x, y)$ is $\sqrt{C(a, b)}$.
- From an external point (a, b) two Tangents can be drawn to the Circle $C(x, y)$, the Equation of the pair of Tangents is $C(x, y) \cdot C(a, b) = T^2(x, y)$ where $T(x, y) = ax + by - r^2$
- The Equation of the Normal to the Circle with center $(-g, -f)$ and on the point (a, b) is written as $y - b = [\,(b+f)/(a+g)\,](x-a)$.

- The Equation of the Chord of Contact is $T(x, y) = 0$.
- If the midpoint of the Chord is (a, b), then the Equation of Chord is $T(x, y) = S(a, b)$.
- If $C_1(x, y) = 0$ and $C_2(x, y) = 0$ are two given Circles, then the family of Circles passing through the point of intersection of the Circles is $C_1(x, y) + \lambda \cdot C_2(x, y) = 0$.
- The angle between two Circles with Radius r_1 and r_2 and d being the distance between Radii is given by $\theta = cos^{-1}\,[\,({r_1}^2 + {r_2}^2 - d)/2r_1r_2\,]$.
- Radical axis of two Circles is $C_1(x, y) = C_2(x, y)$.

Topic - 13: Parabola

- Eccentricity of a Parabola is 1.
- There are four types of Parabolas - Upward, Downward, Leftward and Rightward openings.
- Equation of Rightward Opening Parabola is $y^2 = 4ax$ where a is the distance between the Center and the Vertex of Parabola.
- Equation of Leftward Opening Parabola is $y^2 = -4ax$ where a is the distance between the Center and the Vertex of Parabola.
- Equation of Upward Opening Parabola is $x^2 = 4ay$ where a is the distance between the Center and the Vertex of Parabola.
- Equation of Downward Opening Parabola is $x^2 = -4ay$ where a is the distance between the Center and the Vertex of Parabola.
- Length of Latus Rectum of $y^2 = \pm 4ax$ or $x^2 = \pm 4ay$ is $4a$.
- Parametric representation of $y^2 = 4ax$ is $x = at^2$ and $y = 2at$.
- Define $P(x, y) = y^2 - 4ax$ and any arbitrary point (a, b), then -
  - $P(a, b) = 0 \Rightarrow$ The point is on the boundary of the Parabola.
  - $P(a, b) < 0 \Rightarrow$ The point is inside the Parabola.
  - $P(a, b) > 0 \Rightarrow$ The point is outside the Parabola.
- Equation of a Chord joining the points t_1 and t_2 is $(t_1 + t_2)y = 2x + 2at_1t_2$.
- Equation of Tangent to $y^2 = 4ax$ with Slope m is $y = mx + (a/m)$.
- Equation of Tangent to $x^2 = 4ay$ with Slope m is $y = mx - am^2$.
- Equation of Tangent to $y^2 = 4ax$ at the point (a, b) is $by = 2a(x + a)$.
- Equation of Tangent to $y^2 = 4ax$ at the point t is $ty = x + at^2$.
- Equation of Normal to $y^2 = 4cx$ at the point (a, b) is $y - b = (-b/2c)(x - a)$.
- Equation of Normal to $y^2 = 4ax$ with Slope m is $y = mx - 2am - am^3$.
- Equation of Normal to $y^2 = 4ax$ at the point t is $y + tx = 2at + at^2$.
- Define $P(x, y) = y^2 - 4cx$ and $T(x, y) = by - 2c(x + a)$, the Equation for the pair of Tangents drawn from the point (a, b) is $P(x, y) \cdot P(a, b) = T^2(x, y)$.
- The Equation for the Chord of Contact is $T(x, y) = 0$ where the Tangents to the Parabola is drawn from an external point (a, b).
- Equation of the Chord with midpoint (a, b) is $T(x, y) = P(a, b)$.
- Length of Co-Tangent is $2at^2$ and the length of Co-Normal is $2a$.

Topic - 14: Ellipse

- Equation of a standard Horizontal Ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b$.

- The Eccentricity of the Horizontal Ellipse is $e = \sqrt{1 - \frac{b^2}{a^2}} \in (0,1)$.
- Coordinates of the Foci of a Horizontal Ellipse are $(\pm ae, 0)$.
- Coordinates of the Vertices of a Horizontal Ellipse are $(\pm a, 0)$.
- Coordinates of the Co-Vertices of a Horizontal Ellipse are $(0, \pm b)$
- The equation of Directrix of a Horizontal Ellipse are $x = \pm a/e$.
- Length of the Latus Rectum of a Horizontal Ellipse is $2b^2/a$.
- Equation of the Latus Rectum of a Horizontal Ellipse is $x = \pm ae$.
- Equation of a standard Vertical Ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a < b$.
- The eccentricity of the Vertical Ellipse is $e = \sqrt{1 - \frac{a^2}{b^2}}$.
- Coordinates of the Foci of a Vertical Ellipse are $(0, \pm be)$.
- Coordinates of the Vertices of a Vertical Ellipse are $(0, \pm b)$.
- Coordinates of the Co-Vertices of a Vertical Ellipse are $(\pm a, 0)$.
- The equation of Directrix of a Vertical Ellipse are $y = \pm b/e$.
- Length of the Latus Rectum of a Vertical Ellipse is $2a^2/b$.
- Equation of the Latus Rectum of a Vertical Ellipse is $y = \pm be$.
- Define $E(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$ and an arbitrary point (a, b), then
  - $E > 0 \Rightarrow$ The point is outside the Ellipse.
  - $E = 0 \Rightarrow$ The point is on the boundary of the Ellipse.
  - $E < 0 \Rightarrow$ The point is inside the Ellipse.
- Parametric Form of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x = a \cos\theta$ and $y = b \sin\theta$.
- The Equation of a Chord joining two points α and β of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{x}{a} \cos\frac{\alpha+\beta}{2} + \frac{y}{b} \sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha-\beta}{2}$.
- The Equation of the Tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with Slope m is $y = mx \pm \sqrt{a^2m^2 + b^2}$.
- The Equation of the Tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (p, q) is $\frac{px}{a^2} + \frac{qy}{b^2} = 1$.
- The Equation of the Tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point θ is $\frac{x}{a} \cos\theta + \frac{y}{b} \sin\theta = 1$.
- The Equation of the Normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point (p, q) is $\frac{a^2}{p}x - \frac{b^2}{q} = a^2 - b^2$.
- The Equation of the Normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with Slope m is $y = mx \pm \frac{(a^2-b^2)m}{\sqrt{a^2+b^2m}}$.
- The Equation of the Normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point θ is $ax \sec\theta - by \csc\theta = a^2 - b^2$.

- The Equation for the Chord of Contact for $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $T(x,y)=0$.
- The Equation for the pair of Tangents to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $E(x,y)\cdot E(p,q)=T^2(x,y)$ where the Tangents are drawn from the external point (p,q).
- The Equation of the Director Circle of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $x^2+y^2=a^2+b^2$.
- The Equation of a Chord in $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with mid-point at (p,q) is $T(x,y)=E(p,q)$.
- The point of intersection of two Tangents at the points α *and* β is

$$\left(a\cdot\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}},b\cdot\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$$

Topic - 15: Hyperbola

- The Equation of a standard Horizontal Hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.
- The Eccentricity of a Horizontal Hyperbola is $e=\sqrt{1+\frac{b^2}{a^2}}$.
- The Coordinates of the Foci of a Horizontal Hyperbola are $(\pm ae,0)$.
- The Coordinates of the Vertices of a Horizontal Hyperbola are $(\pm a,0)$.
- The Equation of the Directrix for a Horizontal Hyperbola are $x=\pm a/e$.
- The Equation of the Latus Rectum is $x=\pm ae$.
- The Length of the Latus Rectum is $2b^2/a$.
- The Equation of the Conjugate Hyperbola is $-\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.
- If the Eccentricities of Horizontal and Vertical Hyperbolas are e_1 *and* e_2, then $\frac{1}{{e_1}^2}+\frac{1}{{e_2}^2}=1$.
- The Equation of a Rectangular Hyperbola is $x^2-y^2=a^2$ or $xy=c^2$.
- The Eccentricity of a Rectangular Hyperbola is $\sqrt{2}$.
- The Parametric Form of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $x=a\sec\theta$ and $y=b\tan\theta$.
- Define $H(x,y)=\frac{x^2}{a^2}-\frac{y^2}{b^2}-1$ and an arbitrary point (p,q), then
  - $H(p,q)<0\Rightarrow$ The point lies inside the Hyperbola.
  - $H(p,q)=0\Rightarrow$ The point lies on the boundary of the Hyperbola.
  - $H(p,q)>0\Rightarrow$ The point lies outside the Hyperbola.
- The Equation of the Tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at the point (p,q) is $\frac{px}{a^2}-\frac{qy}{b^2}=1$.
- The Equation of the Tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ with Slope m is $y=mx\pm\sqrt{a^2m^2-b^2}$.
- The Equation of the Tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at the point θ is $\frac{x}{a}\sec\theta-\frac{y}{b}\tan\theta=1$.

- The Equation of the Normal to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (p, q) is $\frac{a^2}{p}x + \frac{b^2}{q}y = a^2 + b^2$.
- The Equation of the Normal to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with Slope m is $y = mx \pm \frac{m(a^2+b^2)}{\sqrt{a^2-b^2m^2}}$.
- The Equation of the Normal to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point θ is $\frac{a}{sec\theta}x + \frac{b}{tan\theta}y = a^2 + b^2$.
- The Equation for the Chord of Contact is $T(x, y) = 0$.
- The Equation for the pairs of Tangents is $H(x, y) \cdot H(p, q) = T^2(x, y)$ where the Tangents to the Hyperbola is drawn from an external point (p, q).
- The Equation of the Chord with midpoint (p, q) is $T(x, y) = H(p, q)$.
- The Equation of the pair of Asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.
- The Equation of the Tangent to a Rectangular Hyperbola at (p, q) is $\frac{x}{p} + \frac{y}{q} = 1$.
- The Parametric Form of Rectangular Hyperbola $xy = c^2$ is $x = ct\ and\ y = c/t$ where $t \neq 0$.
- The Equation of Normal to the Rectangular Hyperbola is $y - (c/t) = t^2(x - ct)$.

Topic - 16: Limits, Continuity and Differentiability

- Sum Rule: $\lim_{x\to p}[\, f(x) + g(x)\,] = \lim_{x\to p} f(x) + \lim_{x\to p} g(x)$
- Difference Rule: $\lim_{x\to p}[\, f(x) - g(x)\,] = \lim_{x\to p} f(x) - \lim_{x\to p} g(x)$
- Product Rule: $\lim_{x\to p} f(x) \cdot g(x) = \lim_{x\to p} f(x) \cdot \lim_{x\to p} g(x)$
- Quotient Rule: $\lim_{x\to p}[\, f(x) \div g(x)\,] = \lim_{x\to p} f(x) \div \lim_{x\to p} g(x)$ provided $\lim_{x\to p} g(x) \neq 0$
- Scalar Multiple Rule: $\lim_{x\to p} k * f(x) = k * \lim_{x\to p} f(x)$
- Power Rule: $\lim_{x\to p}[\, f(x)\,]^k = k[\, \lim_{x\to p} f(x)\,]$
- Composite Function Rule: $\lim_{x\to p} f(g(x)) = f(\lim_{x\to p} g(x))$ provided $f(x)$ exists at $\lim_{x\to p} g(x)$
- Important Limit Properties -
  - $\lim_{x\to 0}[\, sin\, x \div x\,] = 1 = \lim_{x\to 0}[\, tan\, x \div x\,] = \lim_{x\to 0}[\, arctan\, x \div x\,] = \lim_{x\to 0}[\, arcsin\, x \div x\,]$
  - $\lim_{x\to p}[\, (\, x^n - p^n\,) \div (\, x - p\,)\,] = np^{n-1}$
  - $\lim_{x\to 0}[\, ln\, (1 + x) \div x\,] = 1$
  - $\lim_{x\to 0}[\, (\, e^x - 1\,) \div x\,] = 1$ and $\lim_{x\to 0}[\, (\, a^x - 1\,) \div x\,] = ln\, a$ where $a > 0$
  - $\lim_{x\to 0}(1 + x)^{1/x} = e$ and $\lim_{x\to\infty}(1 + 1/x)^x = e$
  - If $\lim_{x\to p} f(x) = 1$ and $\lim_{x\to p} g(x) = \infty$ then $\lim_{x\to p}[\, f(x)\,]^{g(x)} = exp[\, \lim_{x\to p} g(x)\{\, f(x) - 1\,\}\,]$
- L' Hospital Rule: $\lim_{x\to p}[\, f(x) \div g(x)\,] = \lim_{x\to p}[\, f'(x) - g'(x)\,]$
- Missing Point Discontinuity: $\lim_{x\to p} f(x)\ exists, f(p)\ undefined$
- Isolated Point Discontinuity: $\lim_{x\to p} f(x) \neq f(p)$
- Finite Type Discontinuity: $Left - Hand\ Limit \neq Right - Hand\ Limit$
- Infinite Type Discontinuity: Either $Left - Hand\ Limit \to \infty$ or $Right - Hand\ Limit\ \to \infty$
- Oscillatory Type Discontinuity: Limit oscillates between two finite numbers.

- Intermediate Value Theorem: If $f(x)$ is continuous in the interval $[\,a, b\,]$ then $f(c) = K$ for some $c \in (a, b)$ and $K \in (\,f(a), f(b)\,)$.
- Right-hand Derivative: $f(p^+) = \lim_{h \to 0}[\,\{\,f(p+h) - f(p)\,\} \div h\,]$
- Left-Hand Derivative: $f(p^-) = \lim_{h \to 0}[\,\{\,f(p) - f(p-h)\,\} \div h\,]$
- A function is differentiable only if $f(p^+) = f(p^-)$ at a point $x = p$

Topic - 17: Differential Calculus

- Sum Rule: $(f+g)'(x) = f'(x) + g'(x)$
- Difference Rule: $(f-g)'(x) = f'(x) - g'(x)$
- Product Rule: $(fg)'(x) = (f'g)(x) + (fg')(x)$
- Quotient Rule: $(f \div g)(x) = [\,(gf')(x) - (fg')(x)\,] \div g^2(x)$
- Scalar Multiple Rule: $(k * f(x))' = k * f'(x)$
- Chain Rule: $[\,f\{\,g(x)\,\}\,]' = f'\{\,g(x)\,\} * g'(x)$
- Derivative of some common functions -
  - $(\,Constant\,)' = 0$
  - $(\,x^n\,)' = nx^{n-1}$
  - $(\,e^x\,)' = e^x$ and $(\,a^x\,)' = a^x \ln a$ where $a > 0$
  - $(\ln x)' = 1/x$
  - $(\log_p x)' = 1/(x \ln p)$
  - $(\,\sin x\,)' = \cos x$ and $(\,\cos x\,)' = -\sin x$
  - $(\,\tan x\,)' = \sec^2 x$ and $(\,\cot x\,)' = -\csc^2 x$
  - $(\,\sec x\,)' = \tan x \sec x$ and $(\,\csc x\,)' = -\csc x \cot x$
  - $(\,\arcsin x\,)' = 1/\sqrt{1-x^2}$ where $|\,x\,| < 1$
  - $(\,\arccos x\,)' = -1/\sqrt{1-x^2}$ where $|\,x\,| < 1$
  - $(\,\arctan x\,)' = 1/(1+x^2)$
- Logarithmic Differentiation: $y = [\,f(x)\,]^{g(x)}$ then $y' = [\,f(x)\,]^{g(x)}[\,\{\,g(x) \div f(x)\,\} + g'(x) \ln f(x)\,]$ provided $f(x) > 0\ \forall x \in Domain$
- If $f(x, y) = 0$ then $\dfrac{dy}{dx} = -\dfrac{df/dx}{df/dy}$
- If $x = \phi_1(t)\ and\ y = \phi_2(t)$ then $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$
- If $g(x) = f^{-1}(x)\ or\ g(f(x)) = x$ then $f'(x) \cdot g'[\,f(x)\,] = 1$
- Maxima/Minima are of two types - Local and Global
- $f'(p) = 0\ and\ f''(p) < 0$ then $f(x)$ is maximum in the neighborhood of $x = p$
- $f'(p) = 0\ and\ f''(p) > 0$ then $f(x)$ is minimum in the neighborhood of $x = p$

Topic - 18: Tangents and Normals

- Tangent on $y = f(x)$ at the point $x = p$ has the Slope $\left(\dfrac{dy}{dx}\right)_{x=p}$
- Normal on $y = f(x)$ at the point $x = p$ has the Slope $\left(-\dfrac{dx}{dy}\right)_{x=p}$

- Length of Tangent to the curve $y = f(x)$ at the point (x_1, y_1) is $\left| \frac{y_1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2_{x=x_1}}}{\left(\frac{dy}{dx}\right)_{x=x_1}} \right|$
- Length of Sub-Tangent to the curve $y = f(x)$ at the point (x_1, y_1) is $\left| \frac{y_1}{\left(\frac{dy}{dx}\right)_{x=x_1}} \right|$
- Length of Normal to the curve $y = f(x)$ at the point (x_1, y_1) is

$$\left| y_1 \sqrt{1 + \left[\left(\frac{dy}{dx}\right)_{x=x_1}\right]^2} \right|$$

- Length of Subnormal to the curve $y = f(x)$ at the point (x_1, y_1) is $\left| y_1 \left(\frac{dy}{dx}\right)_{x=x_1} \right|$

Topic - 19: Integral Calculus

- Indefinite Integral of some common functions -
  - $\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ where $n \neq -1$
  - $\int \frac{1}{ax+b} \, dx = \frac{ln|\, ax+b \,|}{a} + C$
  - $\int exp\,(ax+b) \, dx = \frac{exp(ax+b)}{a} + C$
  - $\int p^{ax+b} \, dx = \frac{p^{ax+b}}{a \cdot lnp} + C$ where $p > 0$
  - $\int sin\,(ax+b) \, dx = -\frac{cos(ax+b)}{a} + C$
  - $\int cos\,(ax+b) \, dx = \frac{sin(ax+b)}{a} + C$
  - $\int tan\,(ax+b) \, dx = \frac{ln|\, sec(ax+b) \,|}{a} + C$
  - $\int cot\,(ax+b) \, dx = \frac{ln|\, sin(ax+b) \,|}{a} + C$
  - $\int sec^2\,(ax+b) \, dx = \frac{tan(ax+b)}{a} + C$
  - $\int csc^2\,(ax+b) \, dx = -\frac{cot(ax+b)}{a} + C$
  - $\int csc\,(ax+b) \cdot cot\,(ax+b) \, dx = -\frac{csc(ax+b)}{a} + C$
  - $\int sec\,(ax+b) \, dx = \frac{ln|\, sec(ax+b) + tan(ax+b) \,|}{a} + C$
  - $\int csc\,(ax+b) \, dx = \frac{ln|\, csc(x+b) - cot(ax+b) \,|}{a} + C$

- $\int \frac{1}{\sqrt{a^2-x^2}}\, dx = sin^{-1}\frac{x}{a} + C$
- $\int \frac{1}{a^2+x^2}\, dx = \frac{1}{a} tan^{-1}\frac{x}{a} + C$
- $\int \frac{1}{x\sqrt{x^2-1}}\, dx = \frac{1}{a} sec^{-1}\frac{x}{a} + C$
- $\int \frac{1}{\sqrt{a^2+x^2}}\, dx = sinh^{-1}\frac{x}{a} + C$
- $\int \frac{1}{\sqrt{x^2-a^2}}\, dx = cosh^{-1}\frac{x}{a} + C$
- $\int \frac{1}{a^2-x^2}\, dx = \frac{1}{2a} ln\left|\frac{a+x}{a-x}\right| + C$
- $\int \frac{1}{x^2-a^2}\, dx = \frac{1}{2a} ln\left|\frac{x-a}{x+a}\right| + C$
- $\int \sqrt{a^2-x^2}\, dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2} sin^{-1} x + C$
- $\int \sqrt{x^2+a^2}\, dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2} sinh^{-1}\frac{x}{a} + C$
- $\int \sqrt{x^2-a^2}\, dx = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2} cosh^{-1}\frac{x}{a} + C$
- $\int e^{ax} sin\,(bx)\, dx = \frac{e^{ax}}{a^2+b^2}[a\, sin\,(bx) - b\, cos\,(bx)] + C$
- $\int e^{ax} cos\,(bx)\, dx = \frac{e^{ax}}{a^2+b^2}[a\, cos\,(bx) + b\, sin\,(bx)] + C$

- $\int e^x[f(x) + f'(x)] = e^x f(x) + C$
- $\int e^{f(x)}[1 + xf'(x)] = xe^{f(x)} + C$
- $\int f(x)g(x)\, dx = f(x)\int g(x)\, dx - \int [\frac{d}{dx}f(x) \cdot \int g(x)\, dx]\, dx$
- $\int_a^b f(x)\, dx = \int_a^b f(y)\, dy$
- $\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$
- $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$
- $\int_{-a}^a f(x)\, dx = 0$ if $f(x)$ is an Odd Function
- $\int_{-a}^a f(x)\, dx = 2\int_0^a f(x)\, dx$ if $f(x)$ is an Even Function
- King Property $\int_a^b f(x)\, dx = \int_a^b f(a+b-x)\, dx$
- Queen Property $\int_0^{2a} f(x)\, dx = 2\int_0^a f(x)\, dx$ if $f(x)$ is an Even Function otherwise 0

- $\int_0^{nT} f(x)\,dx = n\int_0^{T} f(x)\,dx$ when $f(T+x) = f(x)$
- $\int_x^{T+x} f(x)\,dx = \int_0^{T} f(x)\,dx$ when $f(T+x) = f(x)$
- $\int_{a+nT}^{b+nT} f(x)\,dx = \int_a^{b} f(x)\,dx$ when $f(T+x) = f(x)$
- $\int_{mT}^{nT} f(x)\,dx = (n-m)\int_0^{T} f(x)\,dx$ when $f(T+x) = f(x)$
- Walli's Formula $\int_0^{\pi/2} sin^n\,x \cdot cos^n\,x\,dx = \frac{[\,(n-1)(n-3)...(1\ or\ 2)\,][\,(m-1)(m-3)...(1\ or\ 2)\,]K}{(m+n)(m+n-2)(m+n-4)...(1\ or\ 2)}$ where $K = \pi/2$ when $m, n\ are\ Even$ or $K = 1$ otherwise
- Leibniz-Newton Formula $\frac{d}{dx}\int_{g(x)}^{h(x)} f(x)\,dx = f[h(x)]h'(x) - f[g(x)]g'(x)$
- Limit of Riemann Sum (Left Rule) $\int_a^b f(x)\,dx = (b-a)\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1} f[a + k(\frac{b-a}{n})]$
- Limit of Riemann Sum (Right Rule) $\int_a^b f(x)\,dx = (b-a)\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n} f[a + k(\frac{b-a}{n})]$
- Important Integrals -
  - $\int_0^{\pi/2} ln\,(sin\,x)\,dx = \int_0^{\pi/2} ln\,(cos\,x)\,dx = -\frac{\pi}{2}\,ln\,2$
  - $\int_a^b |x|\,dx = \frac{b-a}{2}$
  - $\int_a^b \frac{|x|}{x}\,dx = |b| - |a|$
- Area under two curves from $x = a\ to\ x = b$ is $\int_a^b [f(x) - g(x)]\,dx$

Topic - 20: Vectors and 3-D Geometry

- Internal Section Formula using position vectors is $\vec{r} = (n\vec{a} + m\vec{b}) \div (n+m)$
- Vector Equation of a Line is $\vec{r} = \vec{a} + t\vec{b}$
- Three points are collinear if $a\vec{x} + b\vec{y} + c\vec{z} = 0$ and $x + y + z = 0$
- Projection of $\vec{a}\ on\ \vec{b}$ is $\vec{a} \cdot \hat{b}$
- $|\,\vec{a} + \vec{b} + \vec{c}\,|^2 = a^2 + b^2 + c^2 + 2(\,\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$
- Lagrange's Identity $|\,\vec{a}\times\vec{b}\,|^2 = a^2b^2 - (\,\vec{a}\cdot\vec{b}\,)^2$
- Area of a Triangle is the magnitude of $(1/2)[\,\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\,]$
- Area of a Quadrilateral with diagonals $\overrightarrow{d_1}\ and\ \overrightarrow{d_2}$ is $A = (1/2)|\,\overrightarrow{d_1}\times\overrightarrow{d_2}\,|$

- Shortest Distance between Skew Lines is $\left|\frac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right|$
- Distance between parallel lines is $\left|\frac{\vec{b}\times(\overrightarrow{a_1}-\overrightarrow{a_2})}{b}\right|$

$$[\,\vec{a}\quad\vec{b}\quad\vec{c}\,] = \vec{a}\cdot(\vec{b}\times\vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$$

- Scalar Triple Product
- Volume of a Tetrahedron is $[\,\vec{a}\ \ \vec{b}\ \ \vec{c}\,]/6$
- Volume of a Parallelepiped is $[\,\vec{a}\ \ \vec{b}\ \ \vec{c}\,]$
- $[\,\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}\,] = 0$ and $[\,\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}\,] = 2[\,\vec{a}\ \ \vec{b}\ \ \vec{c}\,]$
- $[\,\vec{a}\ \ \vec{b}\ \ \vec{c}\,]^2 = [\,\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\,]$
- Vector Triple Product is $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$
- Distance Formula in Space is $d = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$
- Centroid of a Triangle is $\left(\frac{\sum x}{3}, \frac{\sum y}{3}, \frac{\sum z}{3}\right)$
- Equation of a Straight Line in symmetrical form is $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$
- Equation of a Line through two points is $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$
- Vector Equation of a Plane is $(\vec{r}-\vec{a})\cdot\vec{n} = 0$
- Equation of a Plane in Intercept Form is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
- Equation of a plane in Normal Form is $lx + my + nz = p$
- Angle between two planes with normal vectors $\overrightarrow{n_1}$ *and* $\overrightarrow{n_2}$ is $\frac{\overrightarrow{n_1}\cdot\overrightarrow{n_2}}{n_1 * n_2}$
- Angle between a Line and a Plane is $\frac{\vec{n}\cdot\hat{d}}{n * d}$
- Foot of the Perpendicular of a Point on a Plane is given by $\frac{\Delta x}{a} = \frac{\Delta y}{b} = \frac{\Delta z}{c} = -\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$
- Reflection of a Point on a Plane is given by $\frac{\Delta x}{a} = \frac{\Delta y}{b} = \frac{\Delta z}{c} = -2\left(\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}\right)$
- Two lines are coplanar if and only if $\begin{vmatrix} \Delta x & \Delta y & \Delta z \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$
- Perpendicular Distance of a Point from a Plane is $\left|\frac{ax_1+by_1+cz_1}{\sqrt{a^2+b^2+c^2}}\right|$
- Distance between Parallel Planes is $\left|\frac{d_1-d_2}{\sqrt{a^2+b^2+c^2}}\right|$

Topic - 21: Probability and Statistics

- $Probability = Number\ of\ favorable\ outcomes \div Number\ of\ total\ outcomes$
- $P(A) + P(\underline{A}) = 1$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Conditional Probability $P(A|B) = P(A \cap B) \div P(B)$
- Multiplication Theorem $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$
- Bayes' Theorem $P(B|A) = P(B)P(A|B) \div P(A)$
- Binomial Distribution $P(X = x) = \frac{n}{x} p^x q^{n-x}$
- Mean and Variance of Binomial Distribution are np and npq respectively
- Arithmetic Mean is $\underline{x} = \sum_{k=1}^{n} x_k \div n$ for ungrouped discrete data
- Arithmetic Mean is $\underline{x} = \sum_{k=1}^{n} f_k x_k \div \sum_{k=1}^{n} f_k$ when frequency is given
- Mean by Shortcut Method is $\underline{x} = A + \frac{\sum_{k=1}^{n} f_k d_k}{\sum_{k=1}^{n} f_k}$ where $d_k = x_k - A$ and A is the Assumed Mean
- Mean by Step Deviation Method is $\underline{x} = A + (\frac{\sum_{k=1}^{n} f_k u_k}{\sum_{k=1}^{n} f_k})h$ where $u_k = d_k \div h$
- Weighted Mean is $\underline{x} = \frac{\sum_{k=1}^{n} w_k x_k}{\sum_{k=1}^{n} w_k}$
- Combined Mean is given by $\underline{x} = \frac{\sum_{k=1}^{N} n_k \underline{x}_k}{\sum_{k=1}^{N} n_k}$
- Median for ungrouped distribution is
$$\begin{cases} \left(\frac{n+1}{2}\right) th\ term & if\ n = Odd \\ \frac{\left(\frac{n}{2}\right) th\ term + \left(\frac{n}{2}+1\right) th\ term}{2} & if\ n = Even \end{cases}$$
- Median of Grouped Distribution is $l + \frac{(N/2) - F}{f} \cdot h$
- Mode of a Grouped Frequency Distribution is $l + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \cdot h$
- $Mode = 3 * Median - 2 * Mean$
- Mean Deviation is $\sum_{k=1}^{n} |x_k - \underline{x}| \div n$
- Variance for discrete distribution is $\frac{\sum_{k=1}^{n} {x_k}^2}{n} - Mean^2$
- Variance for frequency distribution is $\frac{\sum_{k=1}^{n} f_k {x_k}^2}{N} - Mean^2$
- Standard Deviation is the Principal Square Root of Variance
- Mean Square Deviation is $S^2 = \sum_{k=1}^{n} (x - A)^2 \div n$ or $S^2 = \sum_{k=1}^{n} f_k (x - A)^2 \div \sum_{k=1}^{n} f_k$

Topic - 22: Mathematical Reasoning

- AND $(\wedge)$ is called the Conjunction Operator.
- OR $(\vee)$ is called the Disjunction Operator.

- NOT $(\neg)$ is called the Negation Operator.
- IMPLIES $(\Longrightarrow \text{ or } \rightarrow)$ is called the Conditional Operator.
- IFF $(\Longleftrightarrow \text{ or } \leftrightarrow)$ is called the Bi-Conditional Operator.
- The Truth Table for Conjunction Operator is

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- The Truth Table for Disjunction Operator is

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- The Truth Table for Negation Operator is

p	$\neg p$
T	F
F	T

- The Truth Table for the Conditional Operator is

p	q	$p \rightarrow q \equiv \neg p \vee q$
T	T	T
T	F	F
F	T	T
F	F	T

- The Truth Table for bi-Conditional Operator is

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

- The Bi-Conditional Operator is Equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.
- Two compound statements are logically equivalent if both have the same Truth Table.
- Tautology is the statement that is always true.
- Fallacy is the statement that is always false.
- If the statement is $p \rightarrow q$ then the Converse is $q \rightarrow p$.
- If the statement is $p \rightarrow q$ then the Inverse is $\neg p \rightarrow \neg q$.
- If the statement is $p \rightarrow q$ then the Contrapositive is $\neg q \rightarrow \neg p$.
- $\neg (p \wedge q) \equiv \neg p \vee \neg q$ and $\neg (p \vee q) \equiv \neg p \wedge \neg q$.